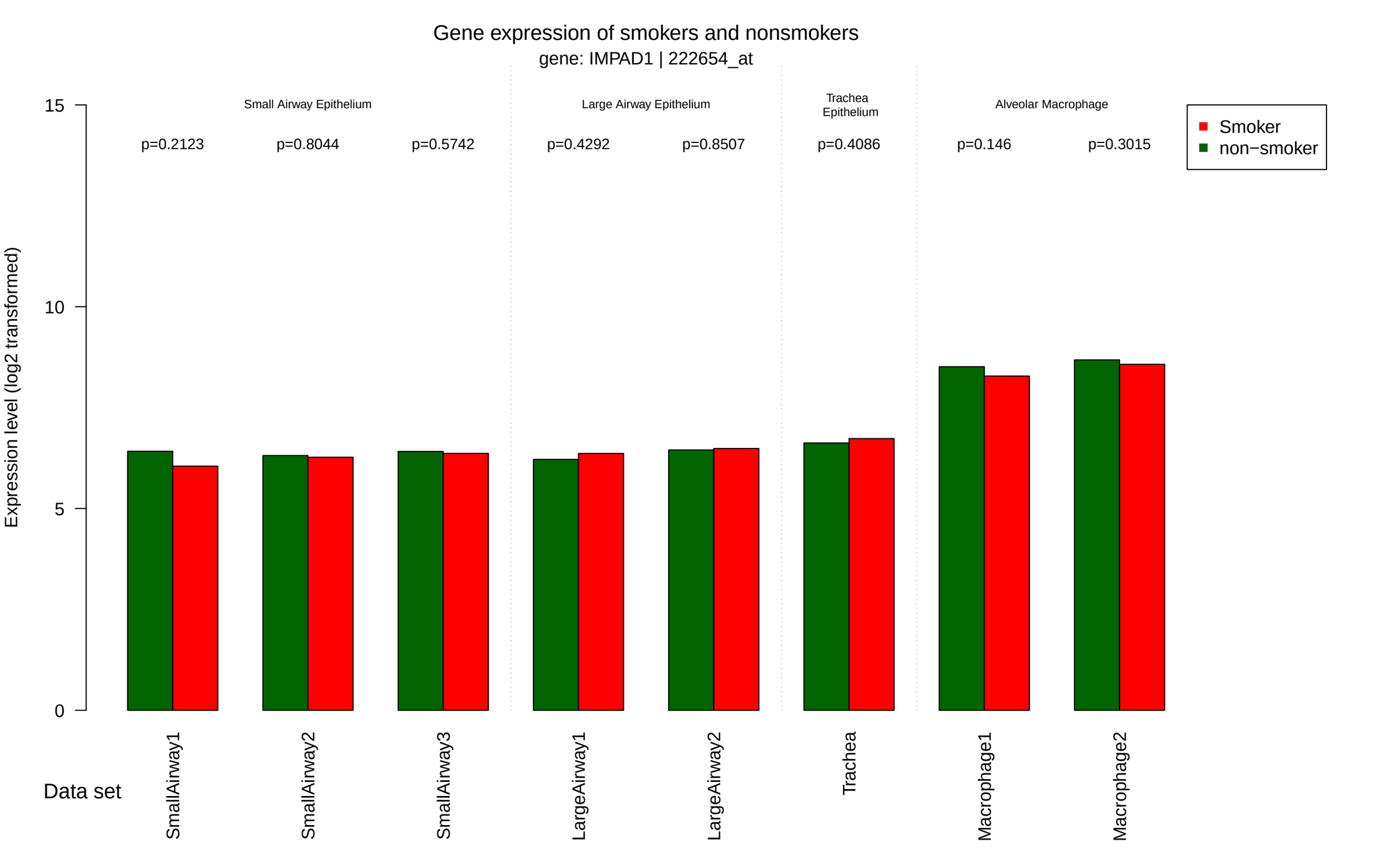
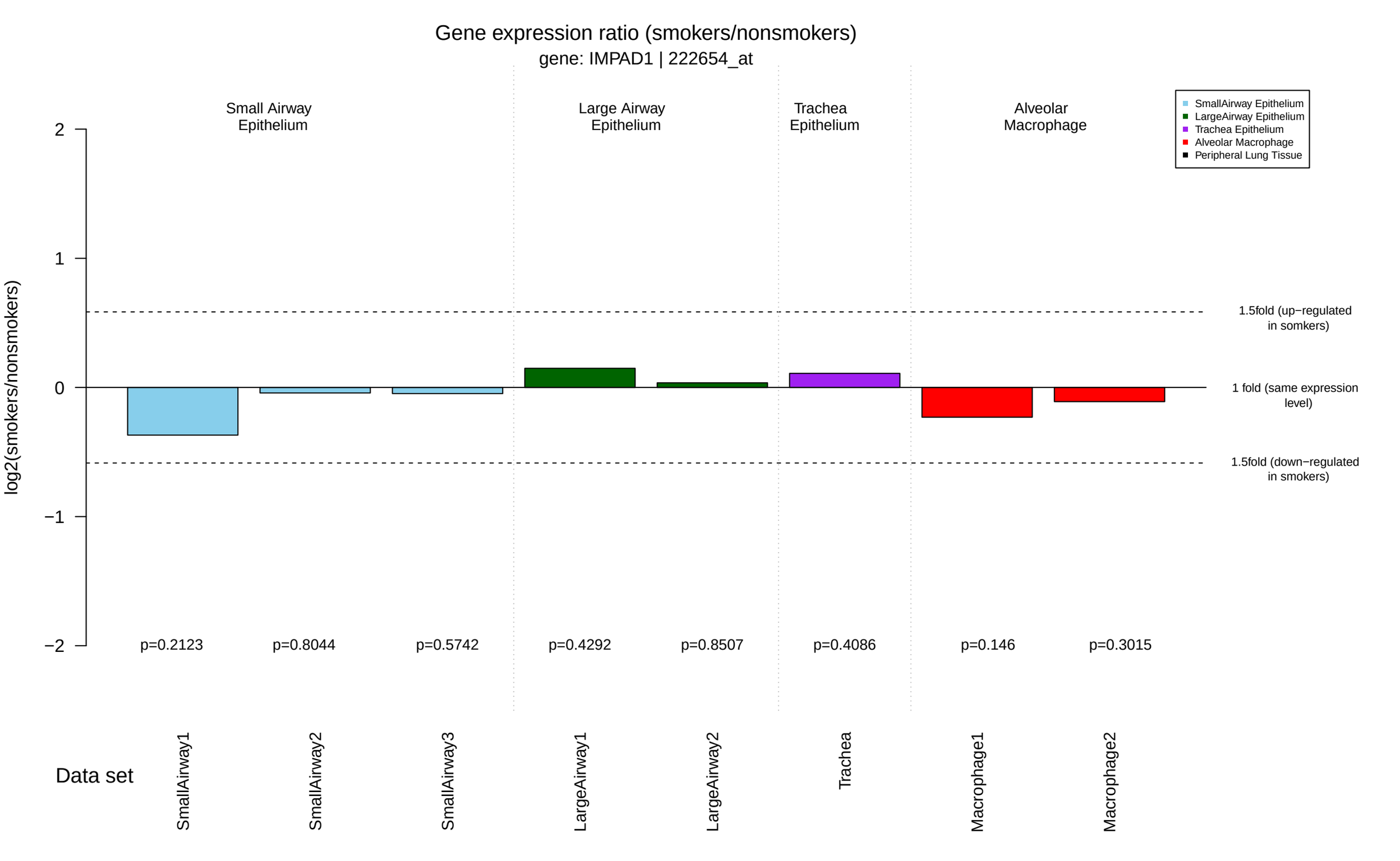
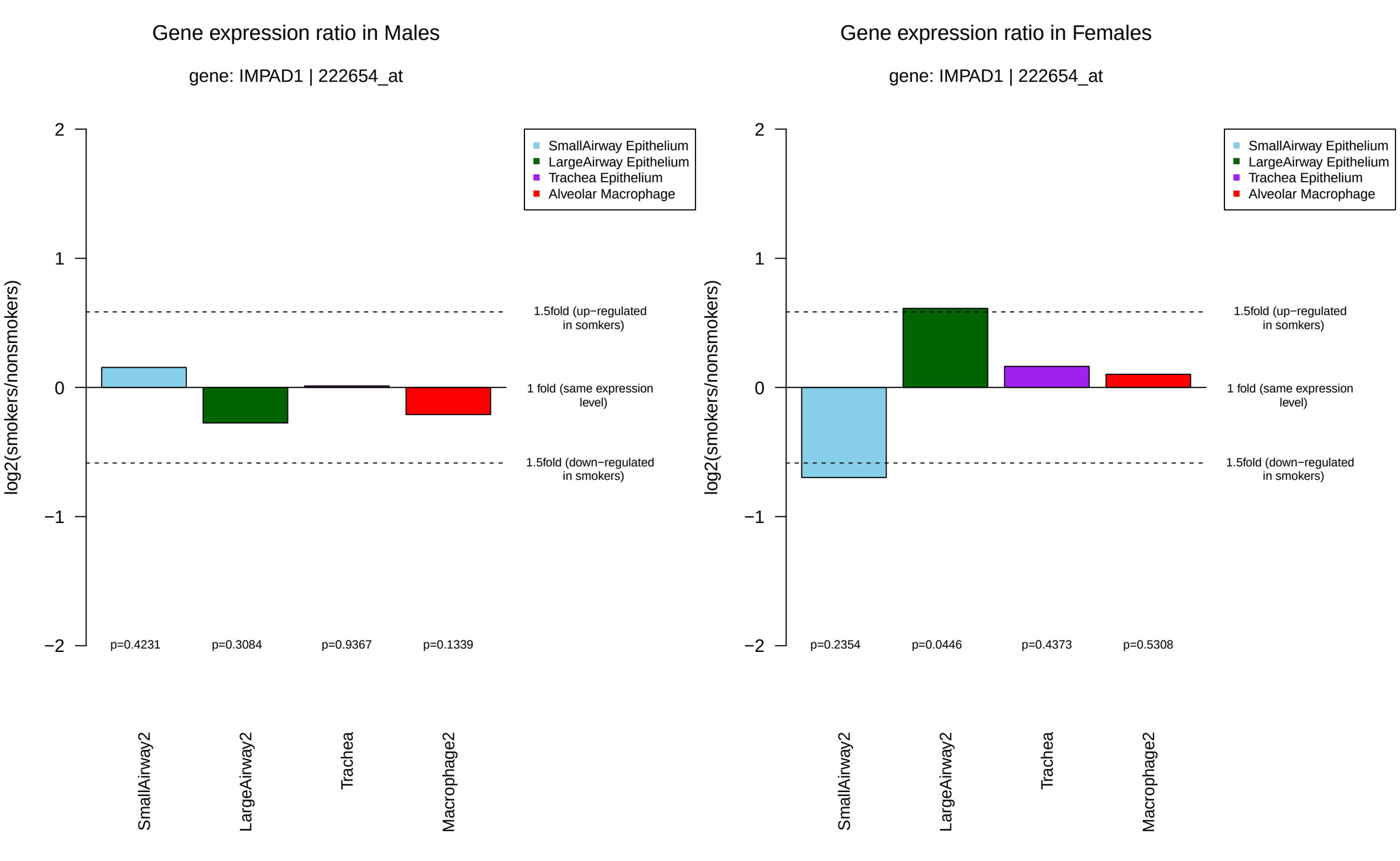
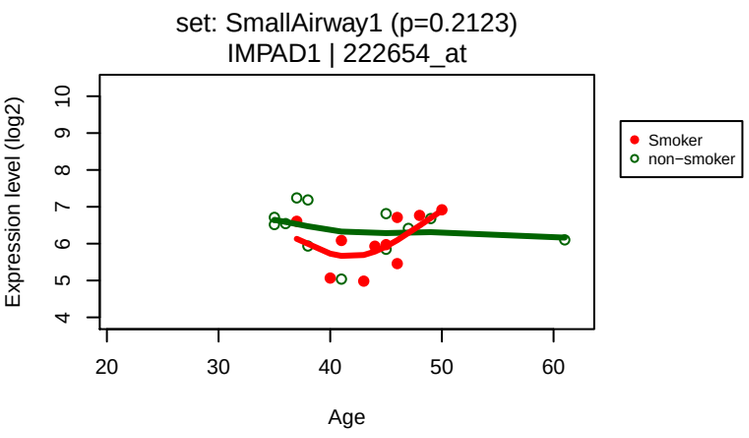
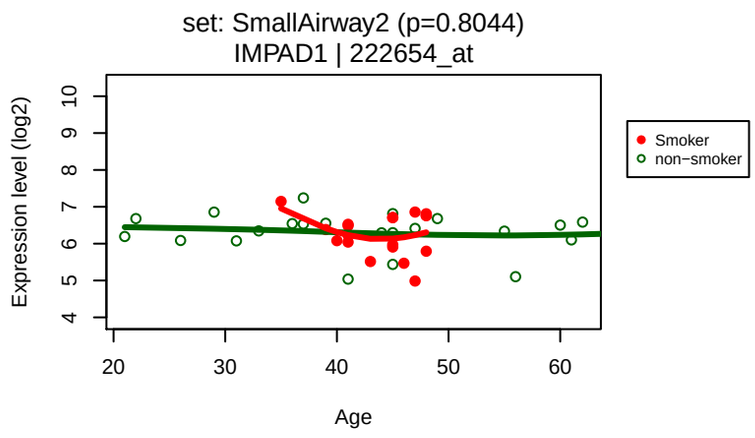
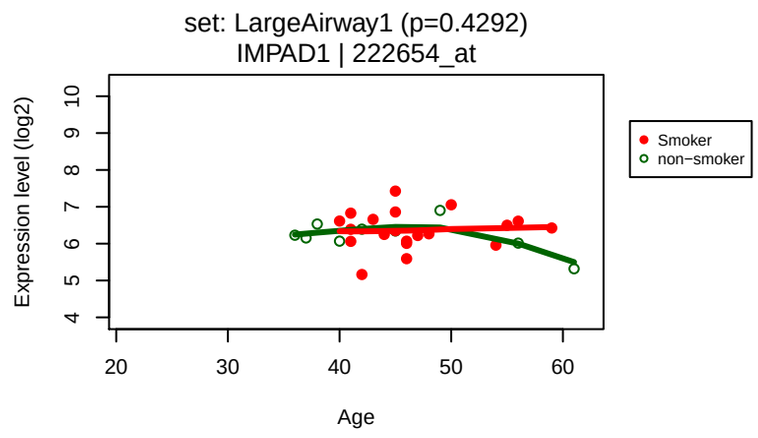

Gene expression of smokers and nonsmokers
gene: IMPAD1 | 222654_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2123
p=0.8044
p=0.5742
p=0.4292
p=0.8507
p=0.4086
p=0.146
p=0.3015
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

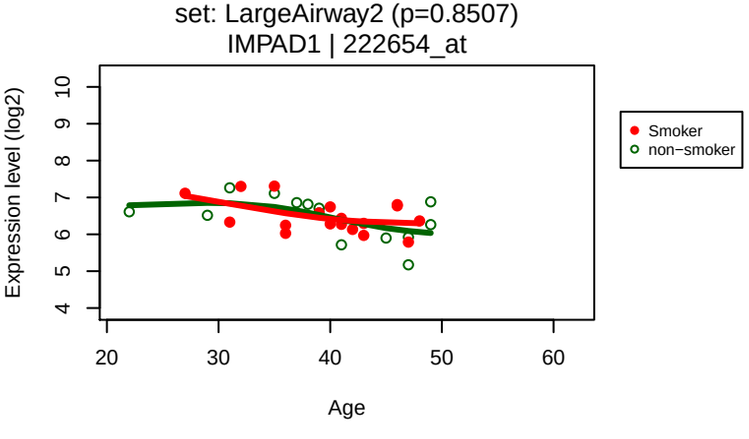

Gene expression ratio (smokers/nonsmokers)
gene: IMPAD1 | 222654_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2123
p=0.8044
p=0.5742
p=0.4292
p=0.8507
p=0.4086
p=0.146
p=0.3015
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

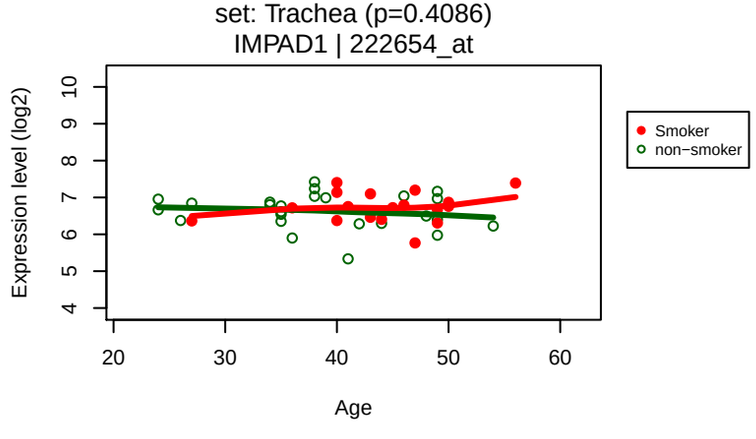

Gene expression ratio in Males
gene: IMPAD1 | 222654_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4231
p=0.3084
p=0.9367
p=0.1339
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IMPAD1 | 222654_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2354
p=0.0446
p=0.4373
p=0.5308
SmallAirway2
LargeAirway2
Trachea
Macrophage2

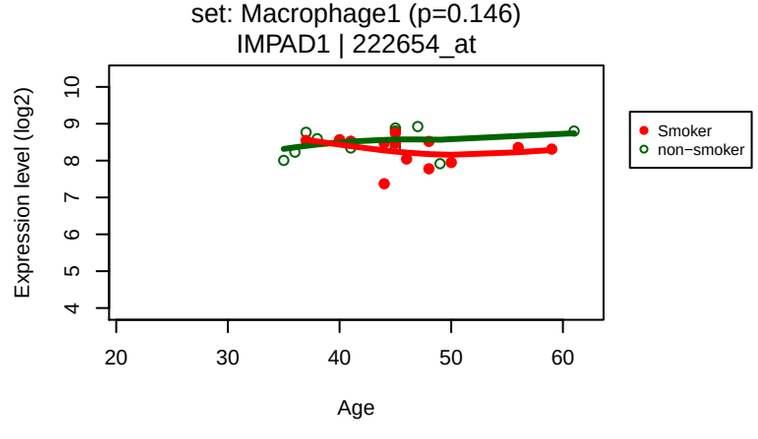

set: SmallAirway1 (p=0.2123)
IMPAD1 | 222654_at
Expression level (log2)
Age
Smoker
non−smoker

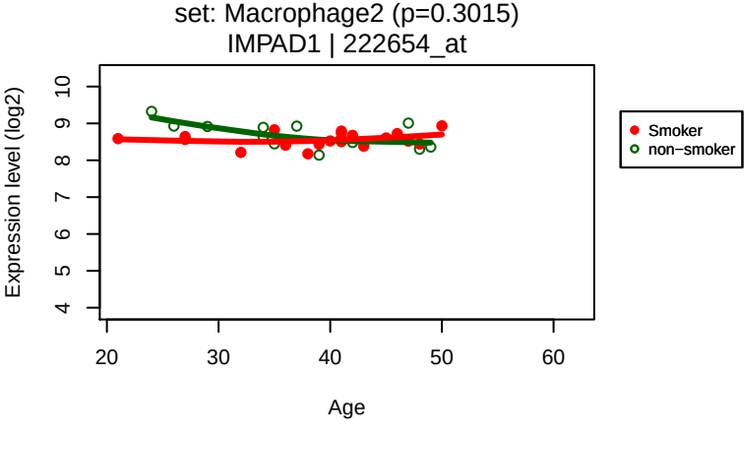

set: SmallAirway2 (p=0.8044)
IMPAD1 | 222654_at
Expression level (log2)
Age
Smoker
non−smoker

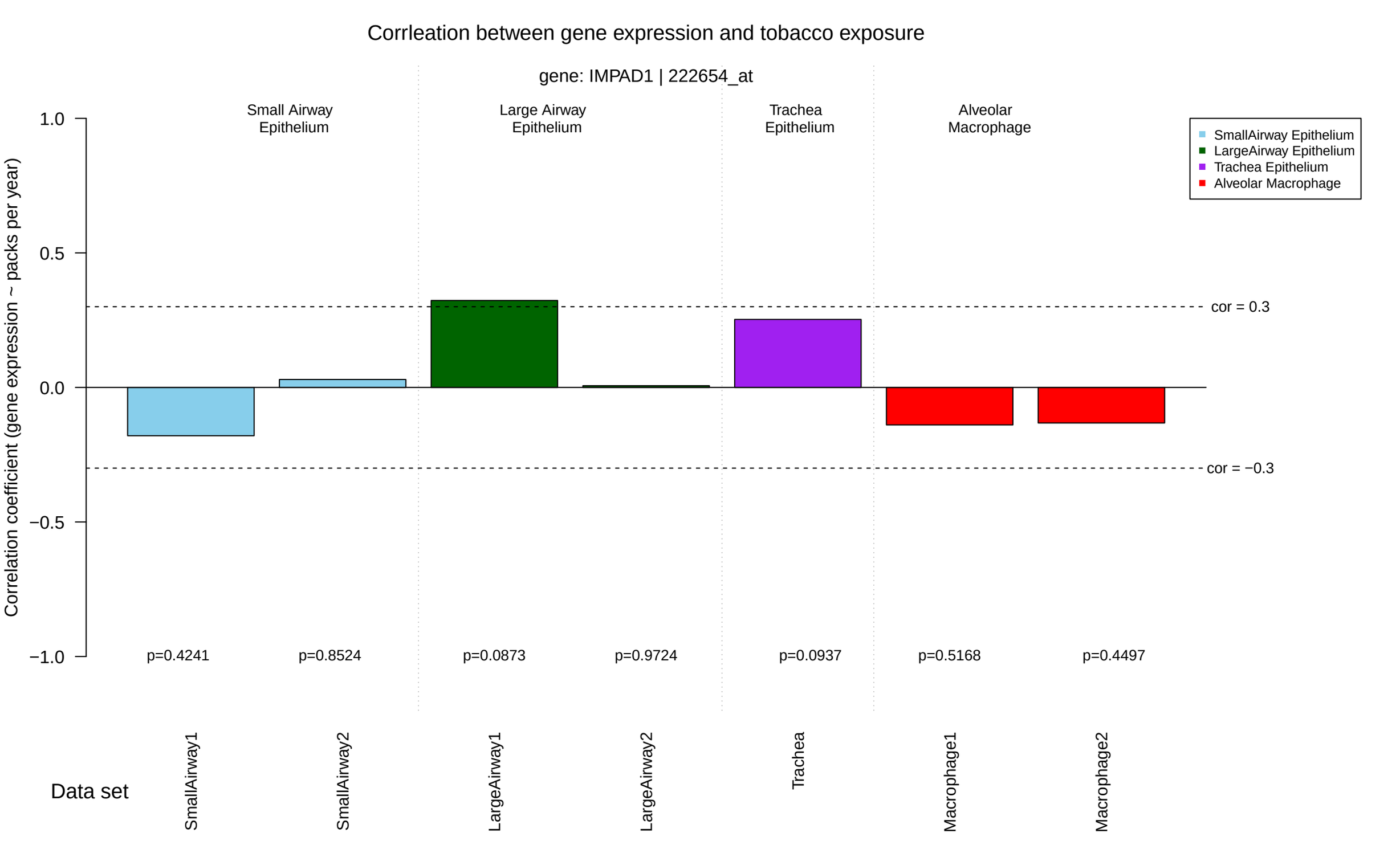

set: LargeAirway1 (p=0.4292)
IMPAD1 | 222654_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8507)
IMPAD1 | 222654_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4086)
IMPAD1 | 222654_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.146)
IMPAD1 | 222654_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3015)
IMPAD1 | 222654_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IMPAD1 | 222654_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4241
p=0.8524
p=0.0873
p=0.9724
p=0.0937
p=0.5168
p=0.4497
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2